



Geometry-based pressure drop prediction in mildly diseased human coronary arteries



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ABSTRACT

Pressure drop (Δp) estimations in human coronary arteries have several important applications, including determination of appropriate boundary conditions for CFD and estimation of fractional flow reserve (FFR). In this study a Δp prediction was made based on geometrical features derived from patient-specific imaging data.

Twenty-two mildly diseased human coronary arteries were imaged with computed tomography and intravascular ultrasound. Each artery was modelled in three consecutive steps: from straight to tapered, to stenosed, to curved model. CFD was performed to compute the additional Δp in each model under steady flow for a wide range of Reynolds numbers. The correlations between the added geometrical complexity and additional Δp were used to compute a predicted Δp . This predicted Δp based on geometry was compared to CFD results.

The mean Δp calculated with CFD was 855 ± 666 Pa. Tapering and curvature added significantly to the total Δp , accounting for $31.4 \pm 19.0\%$ and $18.0 \pm 10.9\%$ respectively at $Re=250$. Using tapering angle, maximum area stenosis and angularity of the centerline, we were able to generate a good estimate for the predicted Δp with a low mean but high standard deviation: average error of 41.1 ± 287.8 Pa at $Re=250$. Furthermore, the predicted Δp was used to accurately estimate FFR ($r=0.93$).

The effect of the geometric features was determined and the pressure drop in mildly diseased human coronary arteries was predicted quickly based solely on geometry. This pressure drop estimation could serve as a boundary condition in CFD to model the impact of distal epicardial vessels.

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1. Introduction

Atherosclerosis is a cardiovascular disease, characterized by plaque buildup in the vessel wall. A subset of atherosclerotic plaques, referred to as vulnerable plaques, contain a necrotic core, covered by a thin fibrous cap often infiltrated by inflammatory cells (Stary et al., 1995). Rupture of the fibrous cap is the main cause of myocardial infarctions (Falk et al., 1995). In previous studies we showed a close association between high wall shear stress (WSS) and plaque weakening in human coronary arteries (Gijsen et al., 2008, 2011). Furthermore cap ruptures are often observed in the upstream region of the plaque (De Weert et al., 2009; Fukumoto et al., 2008; Gijsen et al., 2013) where WSS presumably is high. The combination of high WSS and a thin fibrous cap is therefore potentially a valuable parameter to identify the risk for coronary plaque rupture.

WSS distribution in human coronary arteries can be obtained by combining 3D lumen reconstruction methods with computational

fluid dynamics (CFD) (Kim et al., 2010; Slager et al., 2000; Stone et al., 2012; Van der Giessen et al., 2010). Both lumen reconstruction and CFD are time consuming, especially if a large part of the coronary tree is included in the analysis. However, if WSS would serve as a risk stratification tool to be determined during an intervention, it needs to be calculated rapidly. To reduce the computational time CFD can be restricted to a 3D region of interest, e.g. a coronary bifurcation containing an atherosclerotic plaque. Appropriate boundary conditions should be defined to replace the proximal and distal regions. In steady flow simulations in a bifurcation region, the flow ratio over the side branches highly influences the WSS distribution (Van der Giessen et al., 2011) and it is determined by the pressure drop in the vessels distal to the bifurcation. This pressure drop is composed of an epicardial part and a microvascular part. In this study we focus on the epicardial part. The goal of our study was to develop a pressure drop estimate based on geometry which can be used as a boundary condition to replace the proximal and distal regions in CFD. We chose to investigate mildly diseased vessels, because although they are left untreated, the regions around a plaque are often affected by atherosclerosis as well (Mollet et al., 2005). The lumen of the epicardial coronary tree can be imaged with biplane angiography and reliable 3D lumen reconstructions (Schuurbiers et al., 2009) can be generated in real time (Girasis et

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al., 2011; Tu et al., 2012). This 3D lumen data can potentially serve to estimate the pressure drop in the epicardial vessels.

Real time image based pressure drop estimation in stenosed epicardial vessels can also be useful to determine the fractional flow reserve (FFR). FFR is an important indicator for the hemodynamic significance of a coronary stenosis (Pijls et al., 1996). It is defined as the pressure drop over the stenosis for minimal myocardial resistance. Treatment of the stenosis is warranted if the FFR is below 0.8 (Tonino et al., 2010). During an intervention the FFR is measured with a pressure catheter. With an image based estimate of the pressure drop in a stenosed vessel it is also possible to calculate the FFR instead of measuring it.

Young and Tsai measured the pressure drop over idealized concentric and eccentric stenoses in a straight tube for a wide range of flow conditions (Young and Tsai, 1973a, 1973b). They found a good relationship between the measured pressure drop and the Reynolds number, stenosis length and severity. Gould et al. (1982) applied the relationship to compare the measured pressure drop in coronary arteries to the ones predicted from angiographic data with moderate success. One of the reasons for this is that the relationship developed by Young and Tsai is based on idealized stenoses, while in vivo data show that lesions are much more complex.

This study sets out to find a new estimation of the pressure drop in human coronary arteries using imaging data. The method is based on stepwise adding a complexity to the geometry. The geometrical complexity is increased in three steps, where each additional geometrical feature leads to an increase in pressure drop. The added geometrical feature is subsequently correlated to the additional pressure drop it causes. The image based pressure drop was primarily investigated as a potential boundary condition in CFD, but in this study it was also applied to predict FFR.

2. Material and methods

2.1. Patient selection

We investigated the coronary arteries of 33 patients from a subpopulation of the PROSPECT trial (Clinicaltrial.gov identifier NCT00180466; (Stone et al., 2011)). The patients were treated for an acute myocardial infarction or unstable angina. The coronary arteries were imaged with multi slice computed tomography (MSCT) and intravascular ultrasound (IVUS). Accurate 3D reconstructions of 54 mildly diseased arteries were generated based on a previously developed method (Van der Giessen et al., 2010). Mildly diseased segments were selected because the aim of the study was to investigate regions proximal and distal to a plaque. These regions do not have a significant stenosis but are often affected as well and are considered mildly diseased (Mollet et al., 2005). To ensure a significant pressure drop, 22

segments were selected (9 × LAD, 9 × LCx, 4 × RCA) with a maximum diameter stenosis of 4 × 20–30%, 5 × 30–40%, 10 × 40–50%, 1 × 50–60% and 2 × 60–70%.

2.2. Coronary artery models

Four models of each artery were generated to investigate the effect of the geometrical features on the pressure drop. In each subsequent model a geometrical complexity was added and a feature describing this change in geometrical complexity was correlated to the change in pressure drop it caused (Fig. 1, right panel). First, a straight cylindrical model was created with the radius equal to the inlet radius of the 3D reconstruction. Second, a linearly tapered model was generated. The inlet radius, the outlet radius and the length were based on the full 3D reconstruction. Third, in the stenosed model, the lumen contours obtained with IVUS replaced the lumen of the tapered model. Finally, a curved model was created by adding the centerline obtained with MSCT, which resulted in the complete 3D reconstruction of the artery. The models were generated using in-house developed image processing techniques (Mevislab, Bremen, Germany).

2.3. CFD

The pressure drop in each model was calculated with CFD. A tetrahedral mesh with linear elements was generated for all geometries (Gambit, Ansys, anonsburg, U.S.). Mesh refinement studies resulted in an element size of 1.25×10^{-2} mm. The Navier–Stokes equations were solved using standard numerical techniques (FIDAP, Ansys, Canonsburg, U.S.). The pressure drop was calculated under steady flow conditions for inlet Reynolds numbers (Re) ranging from 5 to 300, with increments of 5 between 5 and 50 and increments of 25 above that. To mimic the coronary flow conditions we extended the inlet with five times the radius (VMTK, www.VMTK.org) and prescribed plug flow. The outlets were stress free. A typical result of the CFD can be seen in Fig. 1 (left panel), which shows the pressure drops at the different Re in the four geometrical models of an artery.

2.4. Modeling the pressure drop

Previous studies already analyzed the typical shape seen in the results of Fig. 1 (Young and Tsai, 1973a). At low flow ($Re < 40$), the pressure loss is a linear function of the average velocity (i.e. the linear domain). At higher flow, the pressure drop is also influenced by convective losses, causing a non-linear pressure drop (i.e. the non-linear domain). Previous studies found that the pressure drop can be described by a second order polynomial

$$\Delta p = \gamma v + \varepsilon v^2 \quad (2.1)$$

here γ and ε are geometry dependent parameters related to the linear and non-linear domain and v the mean velocity at the inlet. If a correlation can be found between these parameters and geometry, the pressure drop in a vessel can be calculated based on geometry.

In addition to Eq. (2.1), an expression for the pressure drop in the linear domain for a tapered vessel can be derived. Similarly to the Poiseuille–Hagen derivation, we assumed that the flow is fully developed and axisymmetric, but now with the boundary conditions

$$v_z = 0 \text{ at } r = \frac{r_0 - r_i}{L} + r_i \quad (2.2)$$

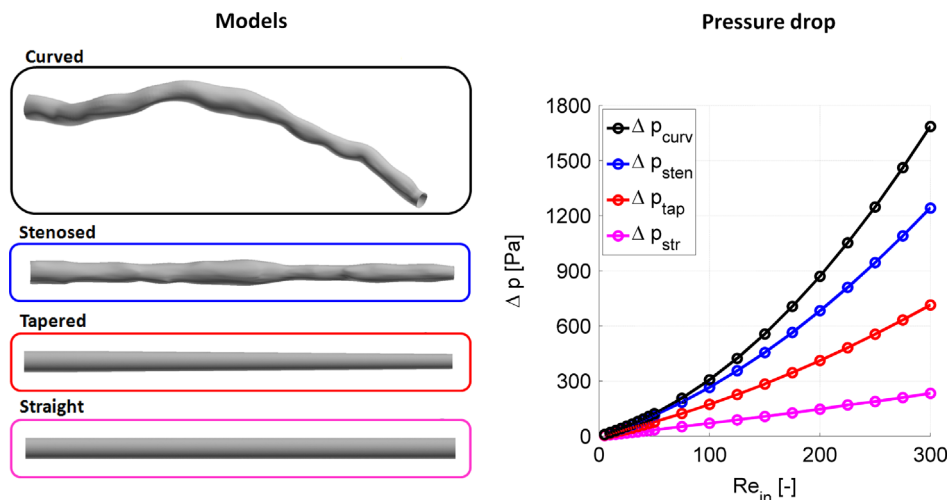


Fig. 1. For each patient specific coronary artery four models with increasing complexity were generated, a straight, tapered, stenosed and curved model (left panel). In each model the pressure drop was calculated using CFD for Reynolds numbers ranging from 0 to 300 (right panel).

$$\frac{\partial v_z}{\partial r} = 0 \text{ at } r = 0 \quad (2.3)$$

where r_i is the inlet radius and r_o the outlet radius. This leads to

$$\Delta p_{\text{tap},0} = \frac{8}{3} \frac{\mu L}{r_o - r_i} \left[\frac{r_o^{-3} - r_i^{-3}}{r_o - r_i} \right] r_i^2 v \quad (2.4)$$

For the derivation it was assumed that there were no radial velocity components. Although the assumptions that lead to the expression are not entirely valid in the tapered model, Eq. (2.4) was used as a first estimate for the pressure drop in the tapered model in the linear domain. The expression was fitted to the CFD result in the linear domain with an additional parameter γ_{tap} . The pressure drop in the non-linear domain in the tapered model was found by fitting ε_{tap} . This gives the expression for the total pressure drop in the tapered model, without having to calculate the pressure drop in a straight tube first

$$\Delta p_{\text{tap}} = \gamma_{\text{tap}} \Delta p_{\text{tap},0} + \varepsilon_{\text{tap}} v^2 \quad (2.5)$$

$$= \gamma_{\text{tap}} \frac{8}{3} \frac{\mu L}{r_o - r_i} \left[\frac{r_o^{-3} - r_i^{-3}}{r_o - r_i} \right] r_i^2 v + \varepsilon_{\text{tap}} v^2 \quad (2.6)$$

Subsequently, we set out to find an expression for the additional pressure drop in the stenosed model. The pressure difference between a tapered and stenosed model was fitted providing the fitting parameters $\gamma_{\text{tap} \rightarrow \text{sten}}$ and $\varepsilon_{\text{tap} \rightarrow \text{sten}}$ in

$$\Delta p_{\text{tap} \rightarrow \text{sten}} = \Delta p_{\text{sten}} - \Delta p_{\text{tap}} \quad (2.7)$$

$$= \gamma_{\text{tap} \rightarrow \text{sten}} v + \varepsilon_{\text{tap} \rightarrow \text{sten}} v^2 \quad (2.8)$$

The same approach was followed to fit the pressure between the stenosed and curved model, resulting in an expression for $\Delta p_{\text{sten} \rightarrow \text{curv}}$. Together this lead to the expression for the total pressure drop which was used to fit Δp for all 22 vessels

$$\Delta p = \gamma_{\text{tap}} \Delta p_{\text{tap},0} + \varepsilon_{\text{tap}} v^2 + \gamma_{\text{tap} \rightarrow \text{sten}} v + \varepsilon_{\text{tap} \rightarrow \text{sten}} v^2 + \gamma_{\text{sten} \rightarrow \text{curv}} v + \varepsilon_{\text{sten} \rightarrow \text{curv}} v^2 \quad (2.9)$$

2.5. Geometric parameters

The next step involved correlating the fitting parameters to the geometrical features of the models. In the tapered model, the only relevant geometrical

parameter was the tapering angle, depicted in Fig. 2a with $\tan(\alpha)$. The stenosed geometry was quantified by three geometrical parameters. If an arterial segment contains a focal stenosis, this might potentially dominate the total pressure drop. We therefore included area stenosis ($AS = \max((A_{\text{tap}} - A_{\text{sten}})/A_{\text{sten}})$). In a more diffusely diseased segment, the gradual changes of the lumen might be a better predictor for the pressure drop. An integral measure of this gradual luminal change can be obtained by computing V_{diff} ($V_{\text{diff}} = V_{\text{sten}} - V_{\text{tap}}$, gray area in Fig. 2b). Finally, we included V_{ratio} to define a geometrical parameter independent of volume of the tapered model ($V_{\text{ratio}} = V_{\text{tap}}/V_{\text{sten}}$).

The curved model was quantified by two parameters. We used tortuosity of the centerline, defined as the ratio of the absolute length between the start and end point of the centerline and the whole length of the centerline. Bijari et al. (2012) used this parameter to study the influence of geometry on flow disturbances in carotid bifurcations. Second, we quantified the curvature by summing the angles between the points along the discretized centerline normalized by the step size: $\Sigma \phi / \Delta s$ (Fig. 2c). This parameter attributes more weight to the local curvature of vessel.

The Pearson r test was used to check for linear regression between the pressure drops and the geometrical features. It is conceivable that combinations of geometrical features influence the resulting pressure drop. Therefore we also correlated cross terms of the quantified geometrical features of subsequent modeling steps to the pressure drops.

The FFR is defined as the ratio between the distal pressure P_d and the aortic pressure P_a under hyperemia. For hyperemic conditions, we used $Re = 250$. Average aortic pressure was assumed to be 90 mmHg (Pijls and Bruyne, 2000). Distal pressure was obtained subtracting the pressure drop from aortic pressure. The pressure drop was derived from CFD to compute the ground truth (FFR_{CFD}) or from Eq. (2.9) to determine $FFR_{\text{estimation}}$.

3. Results

The 3D pressure distribution for three different Re in three arteries is presented in Fig. 3. The coronary arteries in this study included a representative set of mildly diseased segments, with varying length, tapering, degree of stenosis and curvature. As a consequence, a large variation in pressure drops computed with CFD was observed. In general, large pressure drops were observed

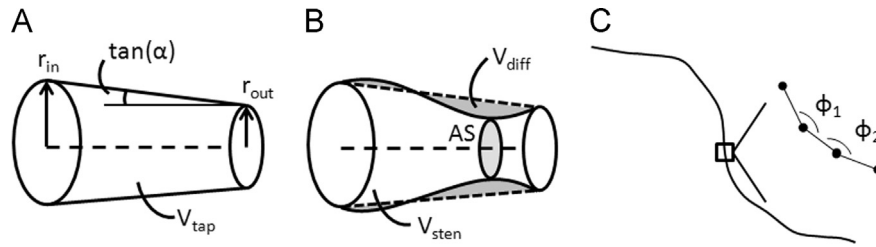


Fig. 2. The geometrical features that were quantified in the three models. (A) The tapered model was quantified with tapering angle α and volume V_{tap} . Here r_i and r_o are the inlet and outlet radius. (B) The stenosed model, where V_{diff} indicates the volume difference (V_{diff}) between the tapered model and the stenosed model (V_{sten}). The maximum degree of stenoses is indicated with AS. (C) The centerline of the curved model was quantified with the angular change from the 3D centerline with the sum of angle of angle ϕ_i .

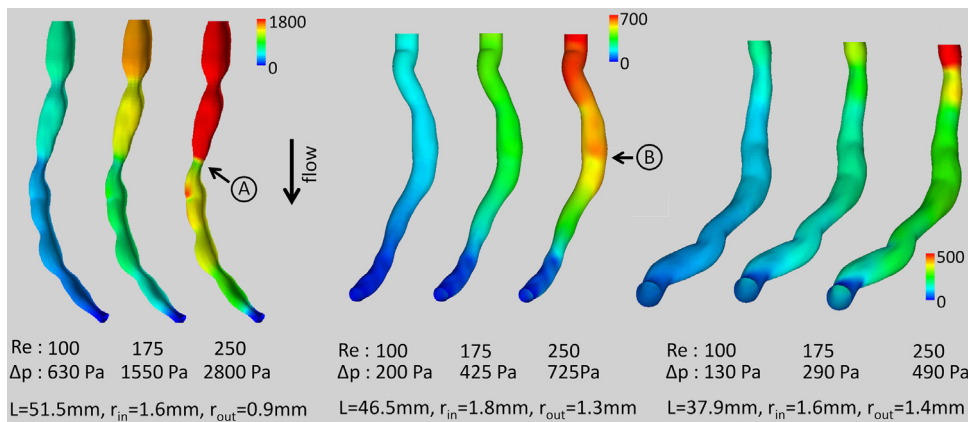


Fig. 3. Three examples of patient specific coronary reconstruction, where the color map represents the pressure drop from CFD. Each artery is depicted at $Re = 100, 175$ and 250 . Large pressure drops mainly occur in stenosed areas (A) and near curves (B). Below the values are given for the pressure drops, length of the vessel and inlet and outlet radius. A large variation in pressure drop and geometry can be seen in the three examples. Note that the color bars were adjusted to optimize visualization of the pressure drop for all Reynolds numbers and the color bars do not cover the full range.

after stenosis (Fig. 3, arrow A) and higher curvature (Fig. 3, arrow B). The pressure drop showed a non-linear increase as a function of the Re .

The mean pressure drop of the 22 coronaries at three representative Re is given in Table 1. The pressure drop shows a non-linear increase as a function of the Re : An increase in the Re of 150% leads to an increase in pressure drop of 300%. The high standard deviation indicates a large variety of pressures drops found for the 22 models. Comparatively high contributions from the straight and tapered model were found. The contributions from the tapered and stenosed models are hardly influenced by the Re , respectively, varying between 31.4% and 32.3% and between 16.1% and 18.7%. With increasing Re , the relative contribution from the straight model decreases from 41.9% at to 31.9%, while the contribution due to curvature increases from 9.7% to 18.0%. At $Re=250$ the contribution of pressure drop in the tapered model is approximately equal to that of the straight tube and the contribution of the curved model is approximately equal to that of the stenosed model.

Results of the fitting and correlating procedures are given in Table 2. In the tapered model the angle was correlated to the pressure drop. For γ_{tap} and ϵ_{tap} a correlation with the tapering angle of $r=0.99$ and $r=0.77$ was found, respectively. In the

stenosed model we found that $\gamma_{\text{tap} \rightarrow \text{sten}}$ had the highest correlation with V_{ratio} ($r=0.85$). For $\epsilon_{\text{tap} \rightarrow \text{sten}}$ the correlation was highest for AS ($r=0.79$). Due to the minimal pressure drop in the linear domain, no significant correlations were found for $\gamma_{\text{sten} \rightarrow \text{curv}}$. For $\epsilon_{\text{sten} \rightarrow \text{curv}}$ we found the strongest correlation with the sum of the angles with $r=0.59$. Additionally, we found one cross terms of two geometrical parameters that had a higher r -value than the separate parameters combined. The product of the tapering angle and AS correlated with ϵ_{sten} with $r=0.89$. This ϵ_{sten} can thus replace ϵ_{tap} plus $\epsilon_{\text{tap} \rightarrow \text{sten}}$ in Eq. (2.9). All the above stated correlations were significant with the Pearson r test for $p < 0.05$. The functions describing the best correlation between the fitting parameters and geometrical features are given in Table 2.

The results from the pressure drop estimation, based on geometry, were compared to the pressure drop found by CFD. The difference between computed and estimated pressure drop as a function of the computed pressure drop is given in Fig. 4 and Table 3. The mean difference in pressure drop is relatively low for all Reynolds numbers, never exceeding 5.4%. However, per vessel considerable relative deviations were observed. At $Re=100$, 90% of the predictions had a relative deviation between -26.0% and 55.7% . At $Re=175$ this was between -29.5% and 60.6% and at $Re=250$ between -31.1% and 64.8% . The symbols in Fig. 4 indicate the maximum degree of stenosis. The symbols are not grouped according to the pressure drop obtained with CFD. This means that the pressure drop is not solely determined by the degree of stenosis but is also by other features, such as tapering and curvature.

The FFR was estimated based on the pressure drop prediction for each vessel. $\text{FFR}_{\text{estimation}}$ showed a high correlation with FFR_{CFD} at hyperemia ($r=0.93$, Fig. 5) and at lower Reynolds numbers ($r=0.95$ at $Re=100$, and $r=0.93$ at $Re=175$). The values for r indicate that the quality of the prediction is independent of the assumed hyperemic flow.

Table 1

On the left the mean total pressure drop at Reynolds number of 100, 175 and 250. The four columns on the right give the mean additional percentage an added geometrical feature had with respect to the total pressure drop.

Re	$\langle P_{\text{CFD}} \rangle$ [Pa (mmHg)]	Straight [%]	Tapered [%]	Stenosed [%]	Curved [%]
100	214 ± 153 (1.6 ± 1.1)	41.9 ± 31.7	32.3 ± 18.5	16.1 ± 22.1	9.7 ± 8.5
175	487 ± 368 (3.7 ± 2.8)	35.8 ± 28.8	31.8 ± 18.8	17.7 ± 20.0	14.8 ± 10.0
250	855 ± 666 (6.4 ± 5.0)	31.9 ± 26.7	31.4 ± 19.0	18.7 ± 18.9	18.0 ± 10.9

Table 2

Overview of correlation between pressure drops and quantified geometrical complexities. The numbers in bold were used in the prediction. The last column shows the found relations between the fitting factors and the geometrical feature.

r	$\tan \alpha$	AS	V_{diff}	V_{ratio}	Tortuosity	$\Sigma \phi_i / \Delta s$	$\text{AS} \cdot \tan \alpha$	Expression
γ_{tap}	0.99							$7.76 (\tan \alpha) + 1$
ϵ_{tap}	0.77							–
$\gamma_{\text{tap} \rightarrow \text{sten}}$		0.76	–0.77	0.85				$18.82 (V_{\text{ratio}}) - 16.62$
$\epsilon_{\text{tap} \rightarrow \text{sten}}$		0.79	–0.37	0.50				–
$\gamma_{\text{sten} \rightarrow \text{curv}}$					–	–		–
$\epsilon_{\text{sten} \rightarrow \text{curv}}$					–0.21	0.59		$2.48 \times 10^{-6} (\Sigma \phi_i / \Delta s) - 0.19$
ϵ_{sten}							0.89	$34.95 (\text{AS} \cdot \tan \alpha) + 0.13$

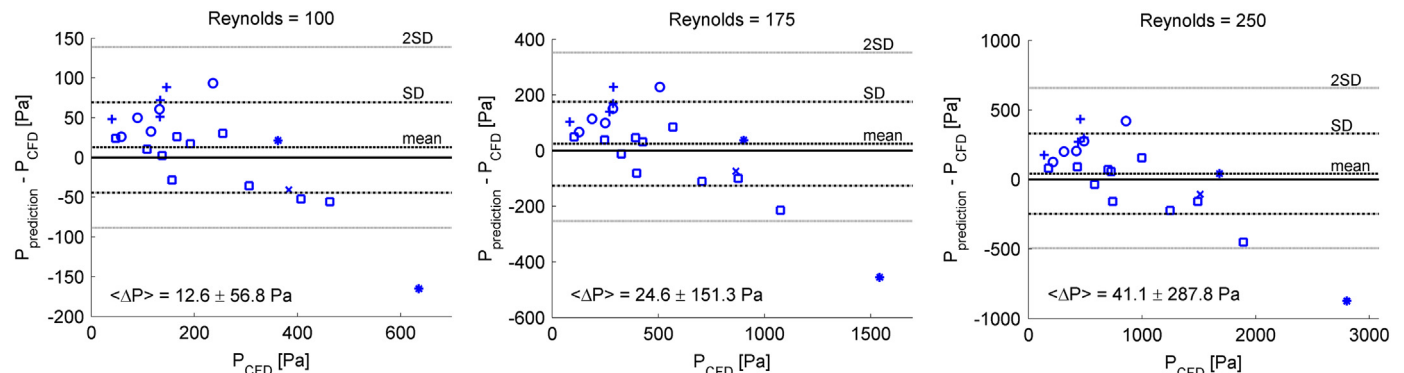


Fig. 4. Difference between the pressure found with CFD and the predicted pressure at Reynolds is 100, 175 and 250. The mean difference at all Reynolds is small and most pressure drop predictions fall within the range of 1 SD of the computed pressure drop. The different symbols indicate the maximum degree of stenosis: + is 20–30%, o is 30–40%, □ is 40–50%, × is 50–60% and * is 60–70%.

Table 3

Mean pressure drop from CFD and the mean difference between the pressure drop from CFD and from the prediction: $\Delta p = p_{\text{pred}} - p_{\text{CFD}}$.

Re	$\langle p_{\text{CFD}} \rangle$ [Pa]	$\langle \Delta p = p_{\text{pred}} - p_{\text{CFD}} \rangle$ [Pa]
100	214 ± 153	12.6 ± 56.8
175	487 ± 368	24.6 ± 151.3
250	855 ± 666	41.1 ± 287.8

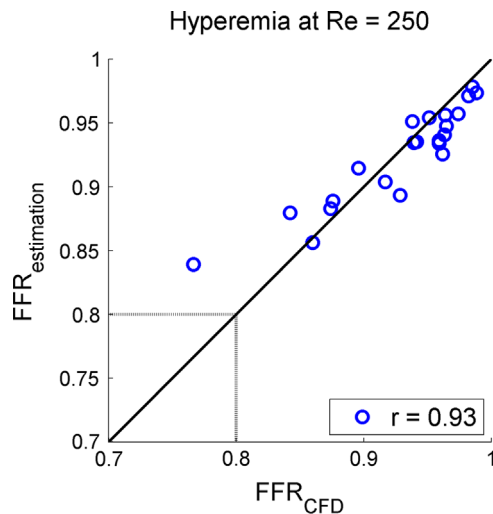


Fig. 5. The correlation between the FFR obtained from CFD and the FFR obtained from our predictions at $Re=250$.

4. Discussion

We used patient-specific geometries to estimate the pressure drop in 22 mildly diseased human coronary arteries. We showed how different geometrical features contribute to the total pressure drop in human coronary arteries. This allowed us to find the correlation between the additional pressure drops and geometric features. This led to an expression that can predict the pressure drop in coronaries based on its geometry.

The 3D patient specific coronary models were deconstructed into four models with increasing complexity. This enabled us to quantify the relative contribution to the total pressure drop for each model separately. When compared to the straight model all other models showed a higher contribution with increasing Re . This can be expected since increased Re cause changes in velocity profiles that generally contain more dissipation and therefore increase the pressure drop in a non-linear fashion. The fact that the relative increase for the tapered and the stenosed model is approximately equal and somewhat lower for the curved model is fully determined by the 3D shape and the degree of stenosis and therefore specific for the data set used in this study. At $Re=250$, the contribution of the tapered model is approximately equal to the straight model and that of curved model approximately equal to the stenosed model. This implies that tapering and 3D curvature, ignored in many previous studies (Banerjee et al., 2008; Van der Horst et al., 2013; Stergiopoulos et al., 1992; Young and Tsai, 1973a, 1973b), should be included when estimating the pressure drop in human coronaries.

Our results show that the additional pressure drop in the tapered model can be estimated very accurately in the linear regime based on Eq. (2.4) and adequately in the non-linear regime. The resulting difference between the computed and estimated pressure drop was only 4.3 ± 15.9 Pa at $Re=100$ and -0.4 ± 111.8 Pa at $Re=250$, indicating that the effect of tapering

on the pressure drop can be accounted for. This result was expected since in this regime the Poiseuille flow approximation is also expected to hold. 3D curvature inherently induces more complex disturbances in the velocity field and it is therefore not surprising that the effect was not captured that adequately with a single geometrical feature.

For the stenosed model we found that the relative pressure drop was mainly influenced by AS, a finding that is in agreement with the findings from Young and Tsai. They calculated the pressure drop based on Eq. (2.1), with γ and ϵ determined by K_v , K_t and a measure for the lumen narrowing. Gould et al. (1982) applied this equation to angiographic images to predict the pressure drop with moderate success in human coronary arteries with a high degree of stenosis. However, due to the high degree of stenosis, different flow regimes occur in these vessels compared to the mildly diseased regions we investigated. They observed a difference between the measured and predicted pressure drop on a vessel to vessel basis of 2500 Pa. Not only were the stenoses not idealized in this study, the aforementioned effects of tapering and 3D curvature were not included either. Despite that our pressure drop prediction shows deviations on a vessel to vessel basis, we did incorporate key geometrical features that determine the pressure drop into the model. This leads us to belief that our model is more suitable for quick pressure drop predictions in mildly diseased regions than predictions based on the models originating from Young and Tsai, since these models were developed for high degrees of stenoses.

The pressure drop prediction in Fig. 4 presents a downward trend. This might be caused by the few cases that present a higher degree of stenosis. As discussed above, regions with a higher degree of stenosis probably present different flow regimes. The model was fitted on mainly mildly diseased cases. The downward trend shows that the model was not able to capture the effects that occur in both the mildly and more heavily diseased regions. A possible solution to capture the effects of the different flow regimes is to use samples from the entire range of stenosis degrees. The model can then be fitted on multiple ranges of stenosis degree.

A high correlation was found between FFR_{CFD} and $FFR_{\text{estimation}}$. Despite the observed discrepancy between computed and estimated pressure drop the relatively high arterial pressure leads to a very good FFR prediction. Another study developed a model to predict FFR with CFD in a reconstruction of the entire coronary tree based on CT data (Kim et al., 2010). This model was applied on large patient populations and the computed FFR was compared to the in vivo measured FFR which served as the gold standard. Moderate to good correlations were found in these studies for a wide range of FFR; $r=0.72$ (Koo et al., 2011) and $r=0.63$ (Min et al., 2012). Whether or not a simpler and faster approach to estimate FFR on imaging data alone can also give valuable results, depends on how well the pressure drop predictions perform in more stenosed segments with lower FFR values. In future studies the clinical relevance of the current findings should be assessed by including more heavily diseased coronary arteries, as well as comparing the predicted FFR to in vivo measurements.

There are several limitations to this study. We selected geometrical features that can be obtained with standard imaging techniques and were thought to cause pressure losses. The possibility that other features, not included in this study, can also be indicative for pressure losses cannot be ruled out. One way to possibly overcome this limitation is to derive a pressure drop estimation from the Navier–Stokes equations. Our good results of the derivation in the tapered geometry illustrate the potential of this approach and could be explored for the stenosed and curved geometries. The same data set was used to first fit the model and subsequently analyze the predicting quality of that model. This approach leads to a minimal average error. In future studies a larger

data set would allow the separation of the training and testing procedure leading to a more thorough analysis of the model. In the reconstructions of the segments, side branches were not included, but this should have no effect on the comparison between CFD result and the prediction. However, when compared to in vivo measurements the effect of tapering can be overestimated.

This method might be used to determine outflow boundary conditions in CFD. However, the approach in this study is only suited for the vessels visible on imaging data. This implies that this pressure drop prediction cannot be applied on vessels of the microvasculature which is important as well. An estimate for pressure drop in the microvasculature needs to be included as well. Other models can be used for this (Huo and Kassab, 2006; Kim et al., 2010; Olufsen, 1999; Pahlevan et al., 2011), although many questions remain about diseased distal beds.

Previous studies used time-averaged WSS to investigate the relation between disease progression and WSS (Stone et al., 2011, 2012; Corban et al., 2014). Results from previous studies suggest that cycle averaged WSS patterns are in good agreement with steady flow patterns (Johnston et al., 2006; Feldman et al., 2002; Myers et al., 2001). Steady flow simulations can thus provide relevant WSS data and are much quicker to perform than transient simulations. Additionally, FFR is calculated as a moving average from pressure data. Therefore this study focused on steady flow. Following from the choice of steady flow simulations, using rigid walls does not influence the outcome in here. Elastic walls only play a role in transient situations where they induce a windkessel effect. In steady simulations this effect cannot occur and therefore the walls can be assumed rigid.

In this study we derived an expression for the pressure drop in mildly diseased coronary arteries based solely on imaging data. On average the predicted pressure drop shows good agreement with the computed pressure drop. The image based pressure drop estimation can be used for quick determination of boundary conditions, paving the way to assess WSS in a clinical setting.

Conflict of interest statement

None declared.

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